

Human-AI Collaboration in Requirements Elicitation: An Empirical Study



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Problem

Alignment issues cause ambiguous, incomplete requirements, delays, and rework.



Context

Elicitation requires stakeholder interaction and clear documentation.



Motivation

Lack of empirical evidence on using AI + collaborative platforms to generate requirements artifacts.

RESEARCH QUESTION

How does an AI-supported collaborative platform influence software requirements elicitation and the production of requirements artifacts compared with alternative elicitation approaches?



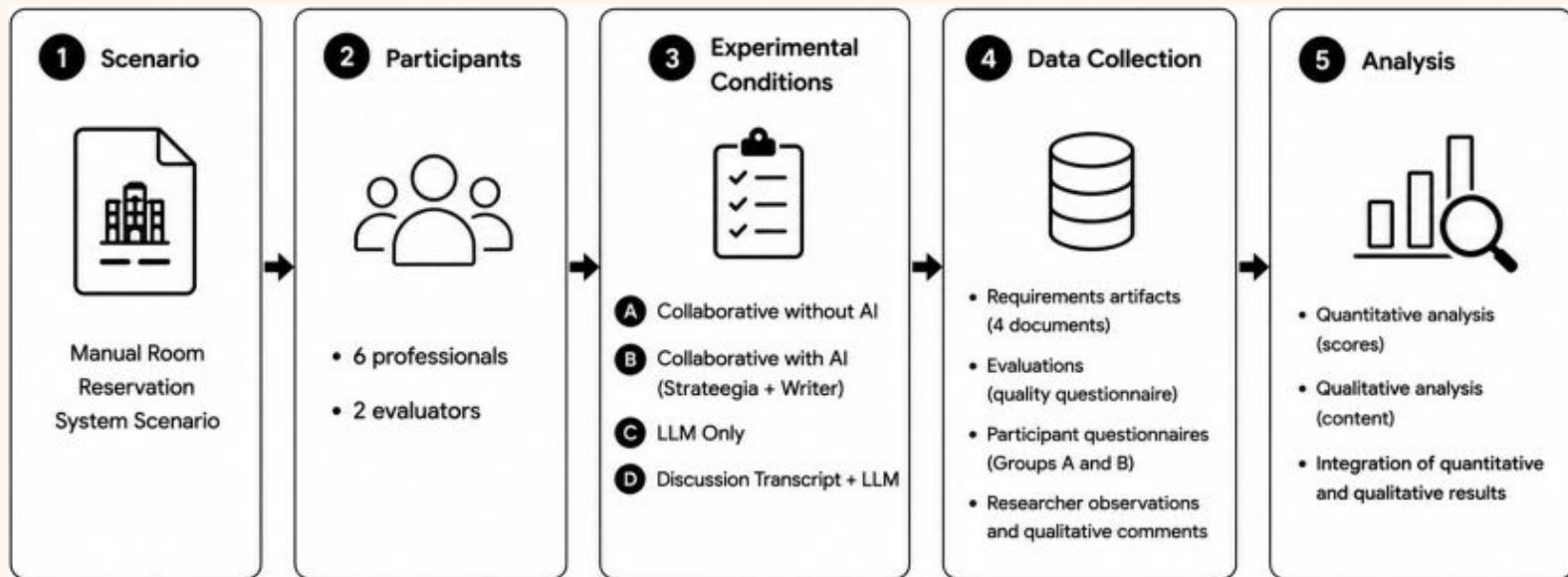
SETUP & CONDITIONS

- **Study:** Between-subjects quasi-experiment using ISO/IEC/IEEE 29148 standards.
- **Scenario:** Fictitious room reservation system.
- **4 Conditions Evaluated:**
 - A: Manual collaboration (No AI)
 - B: Strateegia platform + Writer applet (AI)
 - C: LLM direct prompting (No human discussion)
 - D: Discussion transcript + LLM synthesis

EXECUTION & METRICS

- **Participants:** 2 trios of software professionals composed by:
1 PO, 1 QA and 1 Dev
- **Evaluators:** 2 RE specialist.
- **Metrics:**
 - **Artifact Quality:** Completeness, clarity, consistency, ambiguity, detail, traceability, overall quality.
 - **Process Quality:** Clarity, ease of execution, collaboration, cognitive effort.

METHODOLOGY

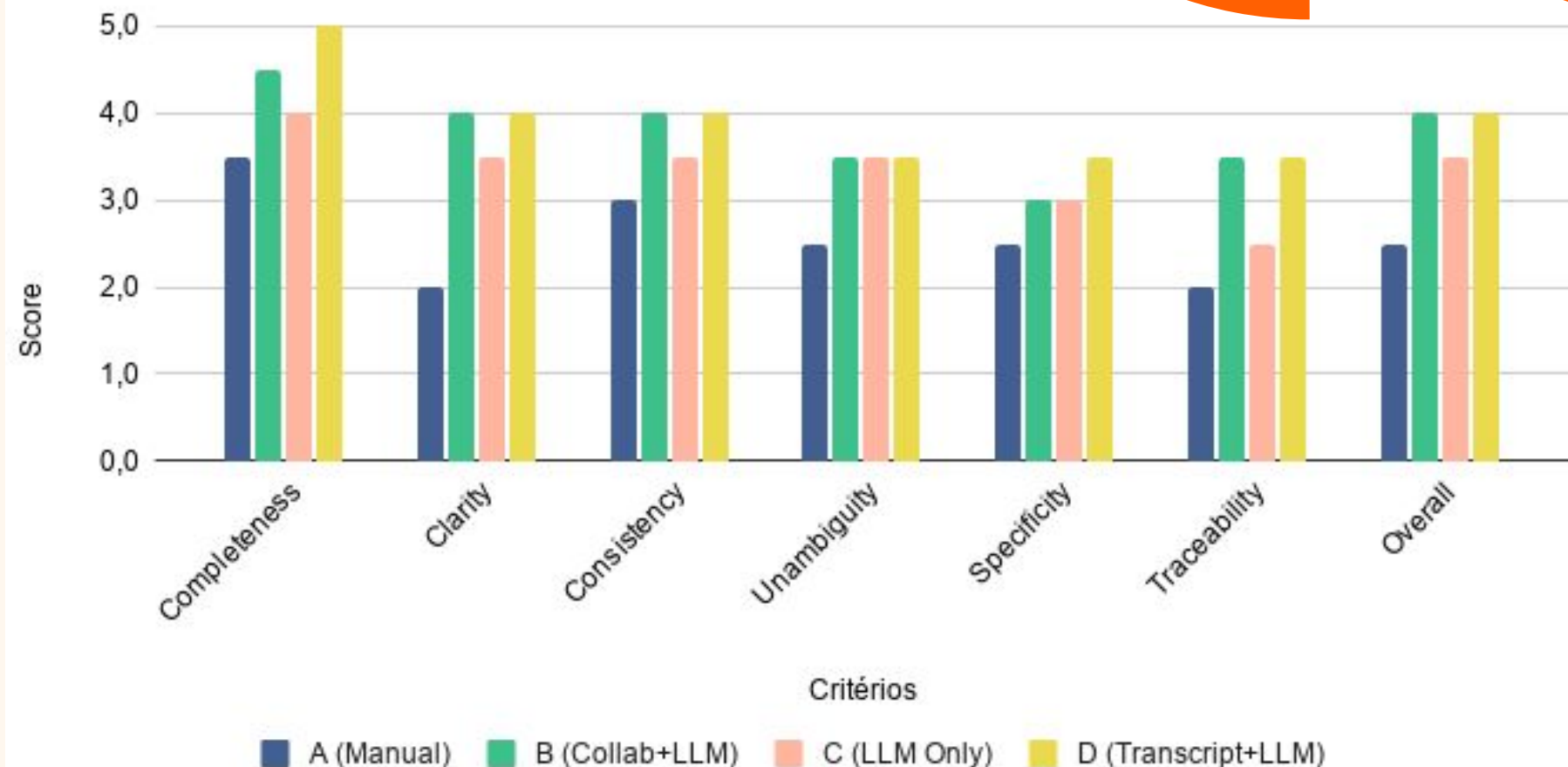


ARTIFACT QUALITY

Criterion	A (Manual)	B (Collab+LLM)	C (LLM Only)	D (Transcript+LLM)
Completeness	3.5	4.5	4.0	5.0
Clarity	2.0	4.0	3.5	4.0
Consistency	3.0	4.0	3.5	4.0
Unambiguity	2.5	3.5	3.5	3.5
Specificity	2.5	3.0	3.0	3,5
Traceability	2.0	3,5	2,5	3,5
Overall Assessment	2,5	4.0	3,5	4.0

Takeaway: Combining stakeholder discussion with AI synthesis (**B & D**) yielded the highest artifact quality.

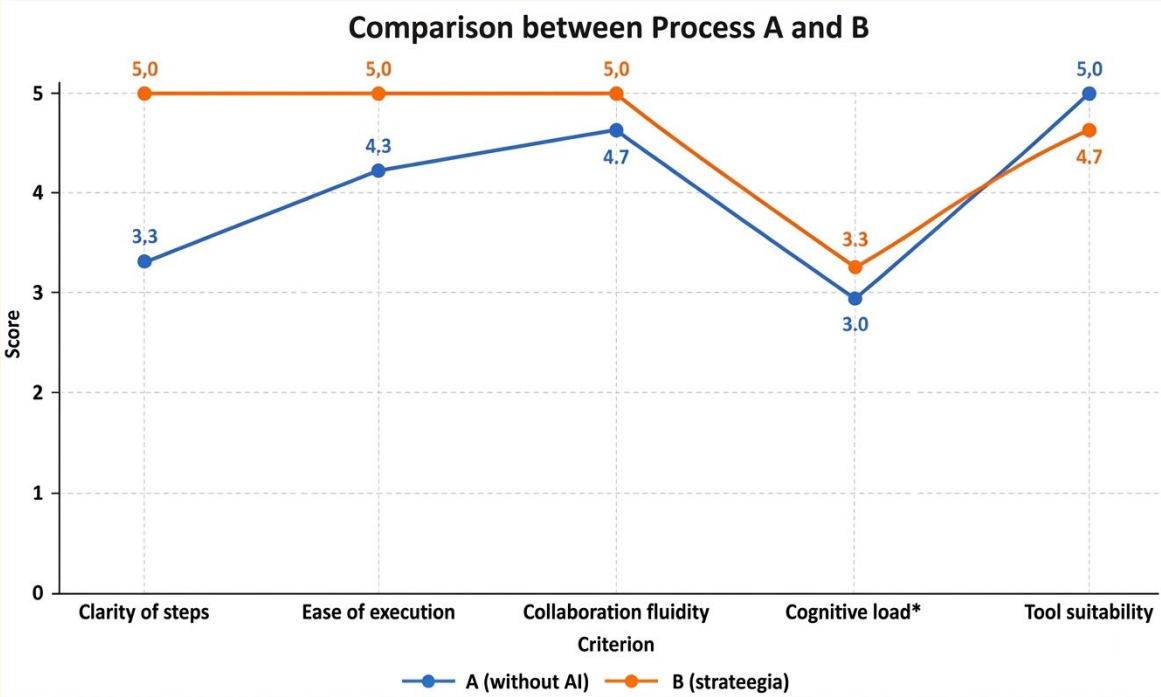
Artifact Quality



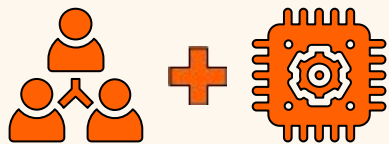
PROCESS QUALITY

Criterion	A	B
Clarity of steps	3.3	5.0
Ease of execution	4.3	5.0
Collaboration fluidity	4.7	5.0
Cognitive load*	3.0	3.3
Tool suitability	5.0	4.7

*In this criterion, unlike the others, higher scores indicate a negative result.



DISCUSSIONS & IMPLICATIONS



Human-AI Synergy

AI should synthesize and document, not replace stakeholder discussion.



Industry Gap

Bridges the gap between unstructured stakeholder debate and formal ISO-compliant artifacts.



CONCLUSION



Findings

AI-assisted synthesis of collaborative discussions produces clearer, more complete, and higher-quality requirements.



Future Work

Test in larger enterprise settings and across broader software domain contexts.





Merci!
Thank you!
Obrigado!

